# MICHAEL G. RIVERA ARROYO

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Scan: GitHub

***Re: Cover Letter · Resume attached***

The following accompanies my resume with a direct note on why its content combines two profiles rarely found in the same person, and why that combination is useful for a serious modern operation.

The first reading of the resume probably raises a reasonable question: why does a field Crew Chief and Instrumentation Specialist with five continuous years of survey work also bring a second technical layer in artificial intelligence, automation, and systems development. The answer is that these are not two separate careers stitched together — they are two sides of the same discipline. Field-grade precision (taking a reading, recording it, verifying it, signing for it) is exactly the same muscle that drives serious software work (version, test, deploy, verify by binary evidence). One feeds the other; they do not compete.

The field layer is direct. More than five years as Crew Chief / Instrumentation Specialist under Javier E. Bidot Associates, PSC — assignments for engineering contractors including Arcadis and DPA, across Puerto Rico and Florida. Thousands of field hours operating Trimble GPS, Trimble Total Station, and the full electromagnetic and ground-penetrating radar locator suite (Ditch Witch, Vermeer G3, Ridgid, US Radar, GSSI). Subsurface utility engineering studies, topographic surveys, blueprint reading, GIS-ready deliverables, bilingual coordination between US-based contractors and Puerto Rico agencies. This is work carried under licensed surveyor supervision with real professional responsibility — not training, and not a transition to something else.

The systems and AI layer is where the proposition becomes distinct. I have been engaging with AI tools since the public release of ChatGPT — first with sustained curiosity, then with growing operational use, and this year I made the decision to treat it seriously as professional infrastructure. The system I describe below was built on top of that trajectory: more than 50 custom command-line tools in Bash and Python, 22 automated processes running continuously on macOS, a local AI stack on my own hardware with no recurring cloud spend (llama.cpp, LanceDB, ONNX), bridges to Anthropic and OpenAI when the work calls for it, authenticated browser automation, and two production client websites deployed 24/7 with a 91.4% external QA pass on the most-audited site. All verifiable in open code on my GitHub profile. This is not aspirational portfolio or demos: it is a system in production used daily.

The center of my proposition — and what differentiates this combination from any standalone "AI consultant" or independent developer — is that the second layer arrives as **FORGE**: an orchestration system directly supervised and operated by a single person. FORGE is the infrastructure that allows me to coordinate multiple artificial-intelligence agents under a doctrine with clear red lines — zero credentials on disk, zero blind automation visible to the client, zero cloud dependency where a local alternative exists, binary evidence of every delivered result. Artificial intelligence maximizes throughput; human supervision sustains professional responsibility and the seal each deliverable carries. In any industry where the result is signed or audited, that distinction matters: AI does not replace the professional or dilute their responsibility — it amplifies them under direct control. It is the difference between delegating a stray script to an opaque tool and operating a codified, versioned, traceable system.

What I offer an organization is that intersection put to work on their real operations: surgical automation of repetitive manual processes, integration between field instrumentation and digital deliverables, processing scripts that compress production time from days to hours, QA and verification pipelines that protect the professional seal, and the capacity to build internal tools fitted exactly to the organization's workflow instead of adopting generic platforms that never quite fit. The conversation I propose is not theoretical: I bring concrete, applicable examples to the first meeting.

I am available for a short conversation at your convenience — on site in the San Juan metropolitan area, hybrid, or remote. If the fit aligns with current priorities, we can define the next step clearly together; if not, the resume sits respectfully in your file.

Thank you for your time and for considering the combination I bring to your operation.

Sincerely,

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